Causal Check on Othello-GPT: Separating Downstream World-Model Computation from Direct Readout Bias

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1 Question, Experimental Setup, and Controls

In 8-layer `Othello-GPT` ($d_{\text{model}} = 512$, `LNPre`), linear probes $W_\ell \in \mathbb{R}^{512\times8\times8\times3}$ decode player-relative board states (EMPTY, YOURS, MINE) with $> 99\%$ accuracy by Layer 4, and adding probe directions across layers steers predictions toward a counterfactual board B' (Nanda et al., 2023). We test two causal questions that arise whenever activation steering alters a model's output:

1. **Mediated world-model computation vs. direct skip-connection bias:** Because the residual stream forms an identity skip connection to the final unembedding ($\mathbf{x}_8 = \mathbf{x}_\ell + \sum_{k=\ell}^{7} f_k(\mathbf{x}_k)$), injecting $\boldsymbol{\delta}_\ell$ across layers $\ell \in \{2..7\}$ also adds $\sum_{\ell=2}^{7} \boldsymbol{\delta}_\ell$ directly to W_U (Figure 1a). Does steering succeed because intermediate Attention/MLP layers recompute Othello flanking rules over B', or merely because $\sum_\ell \boldsymbol{\delta}_\ell$ biases W_U directly?
2. **Decodability vs. single-layer causal control:** Does 99.4% linear probe accuracy at Layer 5 imply that editing Layer 5 alone flips model predictions, or do downstream layers self-repair single-layer edits by attending back to unmodified prefix tokens?

Setup, Metrics, and Controls. We evaluate the official pretrained `synthetic_model.pth` and linear probes (`resid_0..7_linear.pth`) across all $N = 1{,}001$ counterfactual boards in `intervention_benchmark.pkl` (Li et al., 2023; Nanda et al., 2023), where flipping one occupied tile $s = (r, c)$ (YOURS $\leftrightarrow$ MINE) changes ground-truth legal moves from $M(B)$ to $M(B')$. We report 1,000-sample bootstrap 95% CIs on: (1) **Counterfactual Move Error** (↓), $|\hat{M}_K \setminus M(B')| + |M(B') \setminus \hat{M}_K|$ for $K = |M(B')|$ and top-K predictions $\hat{M}_K$; and (2) **Newly-Legal Move Probability Gain** (ΔP_{new} ↑), $\sum_{m\in M(B')\setminus M(B)} [P_{\text{intv}}(m) - P_{\text{clean}}(m)]$. To rule out confounds, we compare: (i) **Mediated-Only** (inject $+\alpha\hat{\mathbf{w}}_s$ at $\ell \in \{2..7\}$, then subtract $-6\alpha\hat{\mathbf{w}}_s$ before $\text{LN}_{\text{final}} + W_U$ so the direct skip path sums to zero) vs. **Direct-Logit Bypass** (leave hidden layers clean and add $+6\alpha\hat{\mathbf{w}}_s$ only before $\text{LN}_{\text{final}} + W_U$); (ii) **Untrained-Encoder Probe** (ridge probes $\lambda = 100$ fit on a randomly initialized 8-layer GPT with matched variance over 200 games); and (iii) **Wrong-Square Control** ($s' \neq s$).

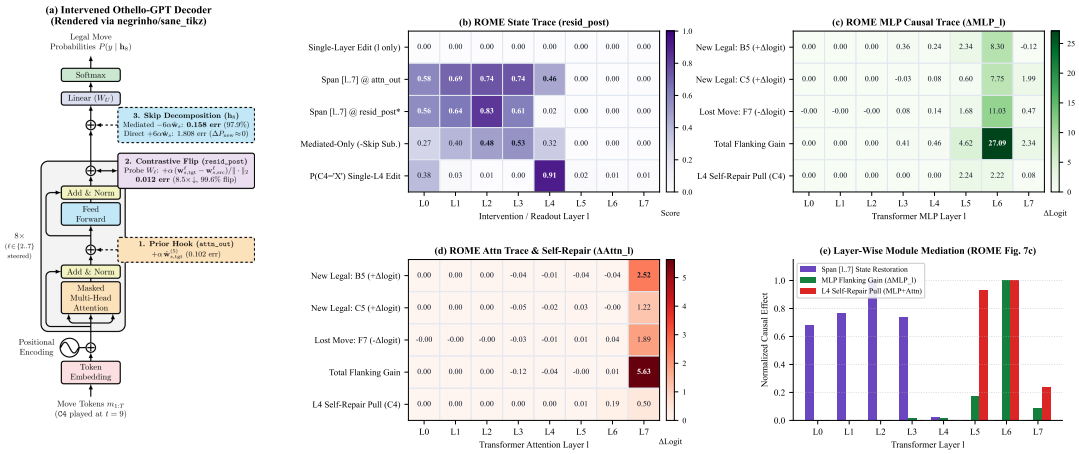


Figure 1: Transformer intervention architecture and ROME causal traces on Case #101 (T=17, flipping C4 from White ◯ to Black ●). (a) Decoder-only `Othello-GPT` diagram rendered via `negrinho/sane_tikz` (`examples/transformer.py`), illustrating the three intervention sites along the residual stream: (1) prior hook at `attn_out` (0.102 err), (2) linear probe W_ℓ and contrastive flip at `resid_post` (**0.012** err, 8.5× ↓), and (3) skip-connection decomposition at $\mathbf{h}_8$ before Linear (W_U) (`Mediated-Only` **0.158** err vs. `Direct Bypass` 1.808 err). **(b–d)** Component-level traces rendered via `kmeng01/rome`'s `plot_trace_heatmap` (`Purples` = state, `Greens` = ΔMLP_ℓ, `Reds` = ΔAttn_ℓ), and **(e)** ROME Figure 7c layer-wise module mediation profile.

Table 1: Causal path decomposition and codebase audit on the official 1,001-board `Othello-GPT` intervention benchmark.

Experimental Arm ($N = 1{,}001$ **counterfactual boards,** $\alpha = 2.3$)	**Error on** B' ↓ **(95% CI)**	ΔP_{new} ↑ **(95% CI)**	**Mechanistic Role / Key Finding**
0. Null Baseline ($\alpha = 0$, un-intervened model)	2.723 [2.645, 2.799]	+0.000 [+0.000, +0.000]	Exact match to Nanda et al. (2023) Table 2 (2.723)
1. Full Multi-Layer Steering (Nanda et al. `intervene.py`)	0.102 [0.072, 0.136]	+0.072 [+0.066, +0.077]	Exact match to Nanda et al. (2023) Table 2 (0.10)
2. **Mediated-Only (Skip-Subtracted before** $\text{LN}_{\text{final}} + W_U$**)**	**0.158** [0.122, 0.198]	**+0.037** [+0.033, +0.041]	**97.9% of error reduction survives zeroing skip path**
3. **Direct-Logit Bypass Control (Only before** $\text{LN}_{\text{final}} + W_U$**)**	1.808 [1.730, 1.884]	**+0.000** [+0.000, +0.000]	**Zero probability gain on newly legal moves** M_{new}
4. Untrained-Encoder Probe Control (Random-init GPT probe)	2.661 [2.581, 2.743]	+0.000 [+0.000, +0.000]	Random-projection directions exert no causal control
5. Wrong-Square Spatial Control (Steer $s' \neq s$ at $\alpha = 2.3$)	3.435 [3.337, 3.530]	+0.001 [+0.001, +0.002]	Editing an unrelated square degrades legal move set $M(B')$
6. Code Audit A: Layer-Matched `probes[`ℓ`]` @ `hook_attn_out`	0.150 [0.116, 0.188]	+0.071 [+0.065, +0.076]	Replaces hardcoded `probes[5]` in `intervene.py`
7. Code Audit B: Layer-Matched `probes[`ℓ`]` @ `hook_resid_post`	0.076 [0.054, 0.102]	+0.075 [+0.069, +0.081]	Aligns intervention hook site with probe training site
8. **Code Audit C: Contrastive Flip** $\hat{\mathbf{d}}_s^\ell$ @ `hook_resid_post`	**0.012** [0.004, 0.026]	**+0.079** [+0.073, +0.085]	8.5× **error reduction** (12 **total errors /** 1,001 **boards**)

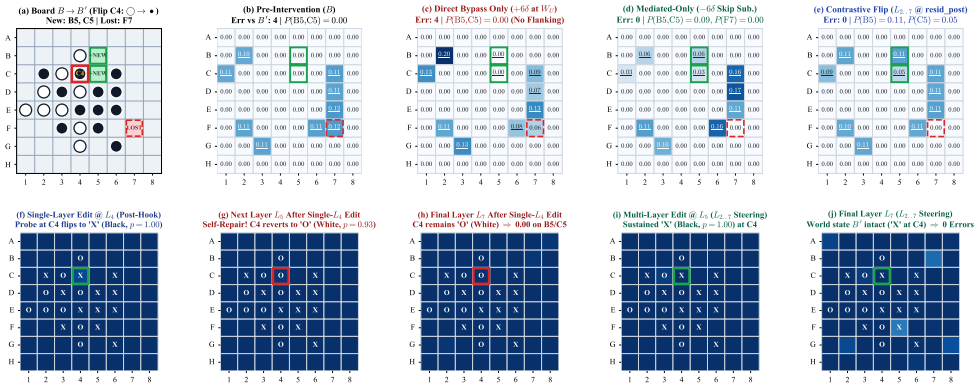


Figure 2: Board-level causal visualization on Case #101 (T=17, flipping `C4`: ○ → ●), rendered via `likenneth/othello_world` (`plot_probs`, `plot_mentals`, `OthelloBoardState`). Top (a–e): Physical transition $B \to B'$ (newly legal moves `B5, C5` in green; invalidated move `F7` in red) and 8×8 move probability heatmaps. `Direct Bypass Only` (c) assigns 0.00 to `B5, C5` (4 errors), whereas `Mediated-Only` (d) and `Contrastive Flip` (e) suppress `F7` ($0.12 \to 0.00$) and activate `B5, C5` with **0 errors**. **Bottom (f–j):** Layer-wise probe beliefs (`O` = White, `X` = Black). A single-layer edit at L_4 flips `C4` to `X` (f), but L_5 and L_7 self-repair `C4` back to `O` (g–h, red box); steering $L_{2..7}$ sustains `X` through L_7 (i–j, green box).

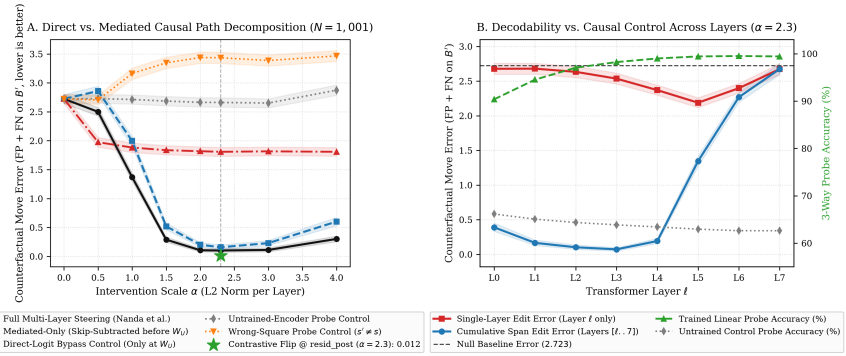


Figure 3: Aggregate results across all $N = 1{,}001$ benchmark boards (95% bootstrap CIs). Left (A): Counterfactual move error vs. scale α. `Mediated-Only` (blue) retains 97.9% of the error reduction, while `Direct-Logit Bypass` (red) fails to discover newly legal moves. Contrastive steering at `resid_post` ($\star$, green) cuts error from 0.102 to 0.012. **Right (B):** Layer-wise 3-way probe accuracy (green/gray, right axis) vs. single-layer edit error (red) and cumulative span $[\ell..7]$ error (blue).

2 Experimental Findings

1. **Downstream world-model computation—not direct readout bias—drives counterfactual steering:** Subtracting the accumulated steering vector before $\mathrm{LN}_{\text{final}}+W_U$ (`Mediated-Only`, Table 1 & Figure 3A) reduces counterfactual move error from 2.723 to $\mathbf{0.158}$, retaining $\mathbf{97.9}\%$ of the full error reduction purely via downstream Attention/MLP computation $\sum_{k=2}^{7} \Delta f_k(\tilde{\mathbf{x}}_k)$ (specifically $\Delta\mathrm{MLP}_5 = +4.62$, $\Delta\mathrm{MLP}_6 = +27.09$, and $\Delta\mathrm{Attn}_7 = +5.63$ logit gain on Case #101; Figure 1c–d). Conversely, applying $+6\alpha\hat{\mathbf{w}}_s$ only before W_U (`Direct-Logit Bypass`) leaves error at 1.808 with $\mathbf{\Delta P}_{\text{new}} = +\mathbf{0.000}$: on Case #101 (Figure 2c), direct unembedding dot products partially suppress invalidated move `F7` ($0.12 \to 0.06$) but assign 0.00 probability to newly legal flanking moves `B5` and `C5`. `Untrained-Encoder` (2.661 err) and `Wrong-Square` (3.435 err) controls confirm spatial and representational specificity.
2. **High single-layer decodability ≠ single-layer causal control due to downstream self-repair:** Trained probe accuracy saturates at 99.0%–99.5% in `L4-L7` (vs. 66.2% → 62.6% on the untrained GPT as random weights scramble token history; Figure 3B), yet single-layer edits never reduce error below 2.188 (`L5`). Figure 1b,e and Figure 2f–h show why: editing `L4` alone flips the probe's belief at `C4` to Black (`X`, $p = 0.91$–1.00), but downstream `L5-L6` MLPs (+2.24, +2.22 repair pull) and `L6-L7` attention (+0.19, +0.50) attend back to unmodified history tokens and overwrite `C4` back to White (`O`, $p = 0.93$). Cumulative spans $[\ell..7]$ achieve minimum error at `Span [L3..L7]` (0.072) and `Span [L2..L7]` (0.102), but fail when started at `L5` (1.345) or `L6` (2.268)—showing that multi-ray flanking circuits in Layers 3–4 execute before linear decodability peaks at Layer 6.
3. **Auditing `ajyl/mech_int_othelloGPT` cuts counterfactual error by 8.5× (0.102 → 0.012):** Inspecting `mech_int/intervene.py` (lines 140–162) revealed three discrepancies between Nanda et al.'s text and code: (i) hard-coding `hook_fns[5]` (`probes[5]`) across all steered layers $\ell \in \{2..7\}$; (ii) hooking `hook_attn_out` (pre-MLP) using probes trained on `hook_resid_post` (post-MLP); and (iii) adding only $+\alpha\hat{\mathbf{w}}_{s,\text{tgt}}$ without subtracting the active source state $\mathbf{w}_{s,\text{src}}$. Aligning the hook to `resid_post` with layer-matched probes drops error to 0.076 (Arm 7), and steering along the unit **contrastive flip vector** $\hat{\mathbf{d}}_s^{\ell} = (\mathbf{w}_{s,\text{tgt}}^{\ell} - \mathbf{w}_{s,\text{src}}^{\ell})/\|\cdot\|_2$ at `resid_post` slashes error to $\mathbf{0.012}$ $[\mathbf{0.004}, \mathbf{0.026}]$ (Arm 8; 12 errors across 1,001 boards, 99.6% probe flip).

3 Falsification Criteria and Next Week's Experiments

- **What would change our conclusion:** (1) Our mediated-computation conclusion would be falsified if freezing intermediate `Attn/MLP` outputs to clean values while applying $\boldsymbol{\delta}$ before W_U recovered $> 50\%$ of the error reduction or $\Delta P_{\text{new}} > +0.02$ (it recovers 33.6% and $\Delta P_{\text{new}} = +0.000$). (2) Our self-repair conclusion would be falsified if projecting out the source tile subspace at `L3` *and* ablating `L4-L6` attention to the move token where s was played still failed to reduce single-layer edit error below 0.50.
- **Next week's plan:** (1) *Sub-circuit path patching of self-repair:* Path-patch individual `L4-L6` attention heads vs. MLP projections to separate attention re-reading token history from MLP local-board heuristics (Lin et al., 2024). (2) *Endgame `MoveFirst` audit ($t > 30$):* Stratify contrastive steering by game turn t and compare discriminative probe normal vectors against difference-in-means vectors ($\boldsymbol{\mu}_{\text{Mine}} - \boldsymbol{\mu}_{\text{Yours}}$) in late-game states where moves are computed before full board state.